Numerical Methods in Context

Exam code: 9MA0

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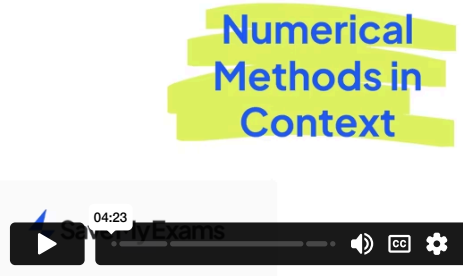
Updated on 16 February 2026



Did this video help you? Yes No

Help me understand

Numerical methods in context

How can numerical methods be used in context?

- **Numerical methods** can be used to model a variety of situations, from the motion of planets across the solar system to the depreciation in a car's value or the fluctuation in crime rates over time
- Numerical methods can be used to find solutions for questions when **analytical methods** may not always work or even be possible
- It is important to understand how each method can be applied to different contexts and the different aspects of a question you may be asked about
- Questions in context will often ask for an **interpretation** or **criticism** of certain parts of the problem

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

FAILURE.
ANY FORMULA OR MODEL CAN BE USED IN CHANGE OF SIGN, BUT SOME MAY NEED COMMON SENSE IDEAS TOO.

a) $f(24) = 48.68$,
$f(25) = -30.96$ (CHANGE OF SIGN)

b) INTERPRETING A SOLUTION MEANS DRAWING ON COMMON SENSE: SOME CARS GET MORE VALUABLE AS THEY GET OLDER

A CARS VALUE IN £'s AFTER x YEARS IS MODELLED BY THE FUNCTION
$f(x) = 17000(0.86)^x - 1000\sin x$
FOR $x > 0$

a) SHOW THAT $f(x)$ HAS A ROOT BETWEEN 24 AND 25

b) EXPLAIN WHY THIS MODEL MAY NOT WORK AS THE CAR GETS OLDER

CHANGE OF SIGN

REPEATEDLY MORE ACCURATE SOLUTIONS– MAY BE ASKED TO REARRANGE EQUATION AND/OR SKETCH DIAGRAM
ITERATION IS COMMON IN REAL LIFE SCIENTIFIC MODELLING.

$x_3 = 43.36$ SO TAKES THREE YEARS FOR ANSWER TO DROP BELOW HALF OF THE ORIGINAL POPULATION

THE ITERATIVE FORMULA
$x_{n+1} = \dfrac{27\sqrt{x_n} + x_n}{0.75e^2}$ IS USED TO MODEL THE DECLINE IN ACTIVE BEE HIVE POPULATION (x MILLION) IN THE UK.
STARTING WITH $x0 = 90$, AFTER HOW MANY YEARS WOULD THE BEE POPULATION MORE THAN HALVE?

ITERATION $x = g(x)$

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CHANGE OF SIGN | **ITERATION $x = g(x)$**

- CONTEXTS COULD INCLUDE TIME, VELOCITY, MOVEMENT OF OBJECTS, MODEL CHANGES IN POPULATION OR THE SPREAD OF DISEASES ETC.
- ALTHOUGH THERE ARE A WIDE NUMBER OF DIFFERENT CONTEXTS, ALL QUESTIONS WILL APPLY THE SAME COMBINATION OF THE ABOVE METHODS.
- SIMPLE IDEAS SUCH AS THE DEPTH OF A STREAM CANNOT GO ABOVE GROUND LEVEL OR A POPULATION CANNNOT GO BELOW ZERO WILL HELP WHEN BEING ASKED TO EXPLAIN THE REASON BEHIND GIVEN INTERVALS OR WHY A ROOT IS PRESENT OR NOT.
- FOR NUMERICAL METHODS, QUESTIONS MAY INVOLVE A GRAPH OR ASK FOR ONE TO BE DRAWN.

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NEWTON–RAPHSON

THE TEMPERATURE θ°C OF A ROOM t HOURS AFTER A HEATING SYSTEM HAS BEEN TURNED ON IS GIVEN BY
$\theta = t + 26 - 20e^{-0.5t}$ WITH $t \geqslant 0$.
STARTING AT $x_0 = 1.9$, USE THE NEWTON–RAPHSON METHOD ONCE TO OBTAIN A SECOND APPROXIMATION.
GIVE YOUR ANSWER TO 3 DECIMAL PLACES.

USING DIFFERENTIATION AND FORMULA TO OBTAIN REPEATEDLY MORE ACCURATE SOLUTIONS – BE CAREFUL OF NEWTON–RAPHSON FAILURE

AS LONG AS THE FUNCTION IS DIFFERENTIABLE AND $f'(x) \neq 0$ THE NEWTON–RAPHSON FORMULA CAN BE APPLIED TO ANY CONTEXT

$f(1.9) = 1.9 + 6 - 20e^{-0.5 \times 1.9}$
$f'(t) = 1 + 10e^{-0.5t}$
$f'(1.9) = 1 + 10e^{-0.5 \times 1.9}$
$\alpha_2 = 1.9 - \dfrac{0.16518}{4.867410} \approx 1.866$

NUMERICAL INTEGRATION
TRAPEZIUM RULE

THE SPEED OF A SMALL BOAT WAS RECORDED EVERY FIVE SECONDS, FROM THE TIME IT LEFT THE JETTY IN A HARBOUR, TO THE TIME IT REACHED OPEN WATER.

TIME (s)	0	5	10	15	20	25
SPEED (ms^{-1})	2	5	10	18	28	42

a) USING THIS INFORMATION ESTIMATE THE LENGTH OF THE HARBOUR FROM JETTY TO OPEN WATER.

b) GIVEN THE BOAT ACCELERATED SMOOTHLY BEFORE REACHING OPEN WATER, EXPLAIN WHETHER YOUR ANSWER TO PART a) IS AN UNDER ESTIMATE OR AN OVER ESTIMATE.

USE TRAPEZOIDAL STRIPS TO CALCULATE THE ESTIMATE FOR THE AREA UNDER A GRAPH – COULD BE OVER OR UNDERESTIMATE AND ASKED FOR PERCENTAGE ERROR.

THE AREA UNDER THE GRAPH CAN REPRESENT PART OF A COMPOUND MEASURE DEPENDING ON THE UNITS.

c) HERE SPEED AND TIME IN

b) INTERPRETING THE SOLUTION CAN BE TRICKY, BUT USING A DIAGRAM CAN HELP SEE WHETHER YOUR TRAPEZIUMS ARE OVER OR UNDERESTIMATE.
GRAPH WOULD BE CONVEX SO OVERESTIMATE

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Examiner Tips and Tricks

- Exam questions will often tell you which method to use
- But beware... numerical methods questions can often combine multiple methods in one large question

Worked Example

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Was this revision note helpful? Yes No

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Chemistry

Physics

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Psychology

All Subjects

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